

A Statistical Thermodynamic Lattice Model for Cu–In Coupled Substitution in Sphalerite

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Abstract

Indium in sphalerite is almost invariably accompanied by copper in a near-1:1 molar ratio, a relation reproduced across volcanogenic massive sulphide, Mississippi Valley-type, vein and skarn deposits and universally attributed to the coupled substitution $2\text{Zn}^{2+} \leftrightarrow \text{Cu}^+ + \text{In}^{3+}$. The relation has been documented extensively but never derived: no existing framework predicts its magnitude, its temperature dependence, or the conditions under which it should fail. We construct a configurational Hamiltonian for the sphalerite cation sublattice, $H = E_c + E_s + E_{sub} + E_{def}$, and sample it by Metropolis Monte Carlo.

An exact topological property of the zinc-blende structure — that two nearest-neighbour cations share precisely one bridging anion — reduces the charge-compensation term without approximation to a nearest-neighbour pair interaction whose coefficient is fixed by the dielectric constant of ZnS rather than fitted, $\lambda = 0.227\text{--}0.368$ eV. The Cu–In pair binding energy is $-2\lambda \approx -0.45$ to -0.73 eV, and the ground state at Cu:In = 1:1 is shown analytically to be exactly the roquesite (CuInS_2) cation ordering, recovered without any input describing that structure.

Simulations at 2 at.% substitution reproduce these predictions. Electrostatics alone enriches Cu–In nearest-neighbour bonds sixteenfold over the random solid solution at 300 °C and removes 96% of the local charge imbalance; heterovalent bonds are enriched 2.7 times more strongly than homovalent ones. Elastic terms, retained at full strength in a control calculation, produce no detectable ordering. The ordering crossover lies at a reduced temperature three to seven times above the entire hydrothermal window, so short-range order is saturated wherever sphalerite forms and carries no thermometric information.

Three consequences follow: indium tenor should be limited by the availability of a monovalent charge compensator rather than by indium supply; vacancy compensation, disfavoured electrostatically by $8\lambda \approx 2.4$ eV, should be diagnostic of copper-poor systems and detectable as cation deficiency; and the ideal dilute-mixing entropy assumed in conventional partition-coefficient models is substantially overestimated. Finite box size prevents resolution of the solid-solution versus nanoinclusion boundary, which we identify as the specific target of large-scale simulation.

Keywords: sphalerite; indium; coupled substitution; charge compensation; statistical thermodynamics; Monte Carlo; roquesite; critical metals

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1 Introduction

1.1 An empirical regularity without a derivation

Among the trace-element relations recorded in sulphide minerals, few are as reproducible as the coupling between copper and indium in sphalerite. Across volcanogenic massive sulphide, Mississippi Valley-type, vein and skarn systems, and over concentration ranges spanning several orders of magnitude, laser-ablation ICP-MS analyses of sphalerite return Cu and In in a molar ratio close to unity (Cook et al., 2009; Frenzel et al., 2016). The relation is sufficiently consistent that departures from it are themselves treated as diagnostic, and it is universally interpreted in terms of the coupled substitution



with a subordinate vacancy-compensated mechanism, $3\text{Zn}^{2+} \leftrightarrow 2\text{In}^{3+} + V_{\text{Zn}}$, invoked where the ratio departs from unity (Johan, 1988; Belissont et al., 2014).

What has not been established is why. Equation (1) is a statement of stoichiometry, not of energetics: it specifies that charge balance must be maintained somewhere in the crystal, but it says nothing about whether Cu and In should occupy adjacent sites or opposite ends of the grain, how strongly they should be associated, how that association should vary with temperature, or under what conditions the alternative mechanism should take over. These are precisely the questions on which the geological utility of the observation depends, and answering them requires a statistical-mechanical treatment of the cation sublattice rather than a balanced chemical equation.

1.2 Why the answer matters beyond mineral chemistry

Indium is classified as a critical raw material in most national assessments, and its supply is unusual in that it is essentially never mined for its own sake: there are no indium ores in the conventional sense, and the overwhelming majority of primary production is recovered as a by-product of zinc refining (Werner et al., 2015; USGS, 2024). The consequence is that the global indium endowment is not controlled by a distinct ore-forming process at all, but by the trace-element chemistry of sphalerite — that is, by the site-occupancy problem posed above. Predicting where indium is concentrated is therefore, quite literally, a problem of predicting the internal configurational state of a mineral.

This inverts the usual relationship between mineral physics and economic geology. For most commodities the mineralogy is a consequence of the ore process and can be treated as a downstream detail; for indium, the mineralogy *is* the ore process. It follows that a predictive theory of trace-element incorporation in sphalerite is not a refinement of exploration practice but a prerequisite for it.

1.3 The theoretical bottleneck

Existing approaches fall into three groups, none of which addresses the configurational question directly.

The first is empirical partitioning. Deposit-type meta-analyses and temperature-calibrated distribution coefficients summarise observed concentrations and their controls with considerable success (Frenzel et al., 2016), but they are descriptive by construction. They cannot predict behaviour outside the calibrated range, and — as shown in Section 5.3 — the ideal dilute-mixing entropy on which such treatments implicitly rest is not a good approximation for a system with the degree of short-range order reported here.

The second is micro-analytical mapping. High-resolution LA-ICP-MS imaging and electron microscopy have established that indium is often distributed heterogeneously within single grains, and have raised the still-unresolved question of whether it resides in genuine solid solution or in nanoscale inclusions of a discrete In phase such as roquesite (Moura et al., 2007). This debate has proved difficult to settle empirically because the two states differ at length scales at the limit of detection. What has been missing is a theoretical criterion predicting which state should occur, and under what conditions.

The third is first-principles defect chemistry. Density-functional calculations of point defects in ZnS and related II–VI semiconductors provide substitution and formation energies of the kind required (Freysoldt et al., 2014; Hoang, 2019), but they characterise isolated or paired defects in periodic supercells. They do not, and are not intended to, describe the statistics of 10^2 – 10^5 mutually interacting substituents at finite temperature. The step from defect energetics to mineral chemistry is a configurational-statistics problem, and it is that step which is absent from the literature.

The gap, stated compactly, is this: the mineralogical community possesses extensive observations of Cu–In coupling and the physics community possesses the energetics of individual defects, but no framework connects the two. This paper is an attempt to supply one.

1.4 Approach and principal findings

We construct a four-term configurational Hamiltonian on the sphalerite cation sublattice,

$$H = E_c + E_s + E_{sub} + E_{def}, \quad (2)$$

in which E_c penalises departures from local electroneutrality within each anion coordination tetrahedron, E_s represents elastic misfit in the Eshelby continuum approximation, E_{sub} carries the chemical cost of substitution referenced to the ore fluid, and E_{def} describes cation vacancies. The Hamiltonian is sampled by Metropolis Monte Carlo with Kawasaki exchange dynamics, and validated against a suite of analytic limits that it reproduces to machine precision.

Two features distinguish the treatment from a phenomenological lattice model. First, the charge-compensation term is not postulated but derived: an exact topological property of the zinc-blende structure allows the tetrahedral electroneutrality functional to be reduced, without approximation, to a composition constant plus a nearest-neighbour pair interaction, whose coefficient is then fixed by the measured dielectric constant of ZnS rather than fitted. Second, the model’s ground state is determined analytically, and proves to be the roquesite cation ordering, obtained without any structural input describing that phase.

The principal findings are:

1. The Cu–In pair binding energy is $-2\lambda \approx -0.45$ to -0.73 eV, an order of magnitude larger than $k_B T$ at hydrothermal temperatures, and contains no adjustable parameter.
2. Charge compensation alone — with all chemical interaction terms switched off — enriches Cu–In nearest-neighbour bonds sixteenfold over the random solid solution at 300 °C and eliminates 96% of the local charge imbalance.
3. Elastic misfit, retained at full strength in a control calculation, produces no detectable ordering. Cu–In coupling in sphalerite is electrostatic, not elastic; the corollary is that Ag–In coupling, with a twentyfold larger size mismatch, should behave qualitatively differently.
4. The ordering crossover lies at a reduced temperature three to seven times above the entire hydrothermal ore-forming window, so short-range order is saturated wherever sphalerite forms and carries no thermometric information.

5. The Cu-compensated substitution is favoured over the vacancy-compensated route by $8\lambda \approx 2.4$ eV in electrostatic energy, yielding a quantitative and falsifiable criterion for the crossover between the two mechanisms.

The calculations are deliberately minimal. They use a $10 \times 10 \times 10$ -cell box at a single composition, and we show explicitly that this is sufficient to establish the ordering physics but insufficient to resolve the solid-solution versus nanoinclusion boundary, because the solute population is exhausted by a single condensed domain. Rather than obscure this limitation we quantify it, and use it to specify the box size, sampling algorithm and parameterisation that addressing the question properly would require.

1.5 Epistemic conventions

Every entry of the parameter table (Table 1) carries one of three designations in its *Basis* column, and each subsection of Section 2 closes with a one-sentence statement of the evidential basis of its content:

- **[F] Established fact.** Crystallographic, spectroscopic or elastic data from the published literature, or an exact mathematical consequence of the zinc-blende topology.
- **[I] Inference.** A quantity derived from [F] inputs through a standard and independently validated physical framework, with no free parameters introduced.
- **[H] Hypothesis.** An original assumption of this work, not yet constrained by measurement or first-principles calculation. Every [H] quantity is either scanned over a range or explicitly flagged as requiring *ab initio* determination.

This convention is retained deliberately: the central claim of the paper is that the observed Cu–In association follows from [F] and [I] alone, and the labelling makes that claim falsifiable rather than rhetorical.

2 Theory

2.1 Nomenclature and configurational variables

Sphalerite crystallises in space group $F\bar{4}3m$ with a cubic cell parameter $a_0 = 5.4093$ Å for the pure ZnS end-member. The cation sublattice is face-centred cubic with coordination numbers $Z_1 = 12$ at $d_{NN} = a_0/\sqrt{2} = 3.825$ Å and $Z_2 = 6$ at a_0 (Shannon, 1976); the sulphur sublattice is a second fcc array displaced by $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4})$, so that every S anion is tetrahedrally coordinated by four cations and every cation by four anions.

We assign to each cation site i a configurational variable $\sigma_i \in \{\text{Zn}, \text{Cu}, \text{In}, \square\}$, where $\square$ denotes a cation vacancy V_{Zn} . The associated formal charges are $q_\alpha = \{2, 1, 3, 0\}$, and we work throughout with the charge deviation from the host cation,

$$\delta q_i \equiv q_{\sigma_i} - q_{\text{Zn}} = \{0, -1, +1, -2\}, \quad (3)$$

so that $\delta q_i = 0$ identically on the undisturbed lattice. A configuration is the complete assignment $\{\sigma\}$ over the N cation sites; there are N anion tetrahedra, indexed by a , and $T(a)$ denotes the set of four cations coordinating anion a .

Basis: the structural data and connectivity are established facts [F]; the configurational variables are definitions.

2.2 A topological lemma

The reduction developed below rests on a single exact property of the zinc-blende connectivity, which is not, to our knowledge, exploited in existing lattice treatments of coupled substitution:

Lemma. In the zinc-blende structure, two cations are nearest neighbours if and only if they share exactly one common anion tetrahedron; second- and higher-neighbour cations share none. Each cation belongs to exactly four tetrahedra.

The lemma follows by direct enumeration of the twelve nearest-neighbour vectors against the four anion offsets, and is asserted computationally over all neighbours of every site in the implementation of Section 3.6. Its consequence is that any energy functional defined on anion tetrahedra maps *exactly*, with no truncation, onto a composition term plus a nearest-neighbour pair interaction on the cation sublattice.

Basis: an exact consequence of the zinc-blende topology [F].

2.3 Local charge compensation, E_c

2.3.1 From screened electrostatics to a local functional

Aliovalent substituents in a semiconducting host interact through a Coulomb potential screened by the dielectric response of the lattice and by any free carriers present:

$$E_c^{\text{Coul}} = \frac{1}{2} \frac{e^2}{4\pi\epsilon_0\epsilon_r} \sum_{i \neq j} \frac{\delta q_i \delta q_j e^{-r_{ij}/\ell_D}}{r_{ij}}. \quad (4)$$

For defect charges in ZnS the screening length is of the order of the nearest-neighbour separation, so the series is dominated by the first coordination shell. Truncating there and invoking the Lemma, we adopt the local electroneutrality functional

$$E_c = \lambda \sum_{a=1}^N (\Delta Q_a)^2, \quad \Delta Q_a = \sum_{i \in T(a)} \delta q_i \quad (5)$$

which penalises departures from charge neutrality within each anion coordination tetrahedron. This is the lattice-discrete expression of Pauling's second rule, and ΔQ_a is the excess of the bond-valence sum reaching anion a over its ideal value of two valence units (Pauling, 1929).

2.3.2 Exact reduction to a pair interaction

Expanding the square and applying the Lemma — each cation appears in four tetrahedra, and each nearest-neighbour pair co-occurs in exactly one tetrahedron where it contributes both orderings — gives the *exact* identity

$$E_c = \underbrace{4\lambda \sum_i \delta q_i^2}_{\text{composition only}} + \underbrace{2\lambda \sum_{\langle ij \rangle} \delta q_i \delta q_j}_{\text{nearest-neighbour pair term}}. \quad (6)$$

Two consequences follow. First, at fixed composition the first term is a constant, so all configurational physics carried by E_c resides in the pair term; the charge-compensation problem is therefore isomorphic to a four-state nearest-neighbour Ising-like model with couplings $J_{\alpha\beta}^c = 2\lambda \delta q_\alpha \delta q_\beta$. Second, the interaction is *strictly* nearest-neighbour: no approximation or cutoff has been imposed beyond the initial truncation of the screened Coulomb series.

2.3.3 A parameter-free estimate of λ

Matching the pair term of Eq. (6) to the nearest-neighbour term of the screened Coulomb series yields

$$\lambda = \frac{e^2}{8\pi\epsilon_0\epsilon_r d_{NN}}. \quad (7)$$

With $e^2/4\pi\epsilon_0 = 14.400 \text{ eV \AA}$ and $d_{NN} = 3.825 \text{ \AA}$, the static and high-frequency dielectric constants of ZnS ($\epsilon_0 = 8.3$, $\epsilon_\infty = 5.13$; Berlincourt et al., 1963; Li, 1984) bracket

$$\lambda \in [0.227, 0.368] \text{ eV}. \quad (8)$$

λ is therefore *not* a fitted parameter. Residual uncertainty enters through the choice of screening length — retaining e^{-d_{NN}/ℓ_D} explicitly would reduce λ by a factor of order e^{-1} for $\ell_D \simeq d_{NN}$ — and through the neglect of Fe 3d states, which modify the dielectric response of natural, Fe-bearing sphalerite. For this reason λ is scanned over $[0, 0.5] \text{ eV}$ in Section 4.4.

2.3.4 Analytical predictions

Equation (6) admits exact results for small defect complexes, all verified numerically in Section 3.6 as regression tests. Writing energies relative to the defect-free lattice:

Configuration	E_c	Origin
Isolated Cu^+ or In^{3+}	4λ	$4\lambda \delta q^2$
Isolated V_{Zn}	16λ	$\delta q = -2$
Nearest-neighbour Cu–In pair	6λ	$8\lambda - 2\lambda$
Ground state of $2\text{Cu} + 2\text{In}$	8λ	four Cu–In bonds
Ground state of $V_{\text{Zn}} + 2\text{In}$	16λ	two $\square$ –In bonds

The binding energy of the elementary coupled-substitution unit is therefore

$$\Delta E_c^{\text{Cu–In}} = -2\lambda \simeq -0.45 \text{ to } -0.73 \text{ eV}, \quad (9)$$

an association energy an order of magnitude larger than $k_B T$ at hydrothermal temperatures ($k_B T = 0.049 \text{ eV}$ at 300°C). The ground state of the four-defect complex, 8λ , lies below that of two independent pairs (12λ) because a configuration in which both Cu are nearest neighbours of both In neutralises four tetrahedra rather than two; clustering beyond the pair is thus electrostatically favoured from the outset, without recourse to any attractive chemical term.

2.3.5 The ground state of E_c is the roquesite ordering

At $x_{\text{Cu}} = x_{\text{In}}$ the charge-compensation term possesses an exact zero-energy ground state: $E_c = 0$ whenever every anion tetrahedron contains two Cu and two In, since then $\Delta Q_a = -1 - 1 + 1 + 1 = 0$ for all a . This is precisely the adamantine cation-ordering rule of chalcopyrite-structured CuInS_2 — the mineral roquesite, which is the observed exsolution product of In-rich sphalerite (Moura et al., 2007). We regard this as the central structural result of the model: *the lowest-energy state of a purely electrostatic local neutrality functional on the zinc-blende cation sublattice is the roquesite structure*, obtained without any input describing that structure.

The ground state is, however, degenerate. Any cation ordering satisfying the $2\text{Cu} + 2\text{In}$ rule attains $E_c = 0$; we verify numerically that both the chalcopyrite-type ordering and a CuAu-I-type layered ordering along $[001]$ do so exactly. Selection among these polytypes must therefore be made by E_s and E_{sub} , and constitutes a well-posed question for first-principles calculation.

2.3.6 Competition between compensation mechanisms

Two charge-balancing routes for In^{3+} are documented in natural sphalerite: monovalent-cation coupling, Eq. (1) (Johan, 1988), and vacancy compensation (Trofimov et al., 2020). Both are contained in the present formalism without additional parameters, because $\delta q_{\square} = -2 = 2\delta q_{\text{Cu}}$. Comparing the two routes at fixed In content, the table of Section 2.3.4 gives $E_c^{(\text{Cu})} = 8\lambda$ against $E_c^{(\square)} = 16\lambda$, an electrostatic preference of $8\lambda \approx 2.4$ eV for the Cu-compensated route. Since the routes differ in atom count, the full thermodynamic comparison requires the reservoir terms of Section 2.5, and the Cu route dominates whenever

$$2\left(\Delta E_{\text{Cu}}^f - \mu_{\text{Cu}}\right) - E_V^f < 8\lambda. \quad (10)$$

Equation (10) is a falsifiable prediction: vacancy compensation should be confined to Cu-poor systems, and the crossover should occur at a Cu activity that the model specifies quantitatively once E_V^f and ΔE_{Cu}^f are known.

Basis: Eqs. (5)–(6) and the tabulated complex energies are exact consequences of the Lemma [F]; the value of λ in Eq. (7), the identification of the E_c ground state with the roquesite ordering, and the criterion of Eq. (10) are parameter-free inferences [I]; the occurrence of both compensation mechanisms in natural sphalerite is an established fact [F].

2.4 Elastic strain energy, E_s

Substituent cations differ in size from the host, and the resulting elastic energy is treated in the Eshelby continuum approximation for a spherical misfitting inclusion in an isotropic matrix (Eshelby, 1957):

$$E_s^{\text{self}} = \sum_i \frac{24\pi \mu_m K_p R^3 \varepsilon_{\sigma_i}^2}{3K_p + 4\mu_m}, \quad \varepsilon_{\alpha} = \frac{r_{\alpha} - r_{\text{Zn}}}{r_{\text{Zn}}}, \quad (11)$$

where $R = (3\Omega_0/4\pi)^{1/3} = 2.11$ Å is the Wigner–Seitz radius of a cation site ($\Omega_0 = a_0^3/4$), and $K = 77$ GPa, $\mu = 30.7$ GPa are the bulk and Voigt–Reuss–Hill shear moduli of ZnS (Berlincourt et al., 1963). Strain-mediated coupling between substituents is represented at leading (elastic-dipole) order by a pair term decaying as r^{-3} ,

$$E_s^{\text{pair}} = \Gamma \sum_{i < j} \varepsilon_{\sigma_i} \varepsilon_{\sigma_j} \left(\frac{a_0}{r_{ij}} \right)^3, \quad (12)$$

truncated at the second coordination shell, so that $\Gamma_2 = \Gamma_1/2^{3/2}$. The sign structure of Eq. (12) encodes strain compensation: substituents of opposite misfit attract, those of like misfit repel.

Inserting Shannon effective ionic radii for fourfold coordination (Shannon, 1976), $r_{\text{Zn}^{2+}} = r_{\text{Cu}^{+}} = 0.60$ Å and $r_{\text{In}^{3+}} = 0.62$ Å, gives $\varepsilon_{\text{Cu}} = 0.000$ and $\varepsilon_{\text{In}} = +0.033$, whence $E_s^{\text{self}}(\text{In}) = 0.033$ eV. Comparison with Eq. (9) gives

$$\frac{E_s^{\text{self}}(\text{In})}{\left| \Delta E_c^{\text{Cu-In}} \right|} \approx 0.05, \quad (13)$$

which motivates the first original hypothesis of this work:

Hypothesis H1. Cu–In coupled substitution in sphalerite is *electrostatically* rather than *elastically* driven, with the strain contribution below 10% of the charge-compensation energy. The corollary is that Ag–In coupling, for which $\varepsilon_{\text{Ag}^{+}} = +0.67$ and the linear-elastic treatment itself becomes marginal, should behave qualitatively differently and should depart from strict 1:1 stoichiometry.

H1 is falsifiable by first-principles substitution energies, and we test its internal consistency in Section 4.1 by comparing the full model against a calculation in which the chemical and elastic terms are switched off entirely.

Basis: the ionic radii and elastic moduli are measured quantities [F] and Eqs. (11)–(12) apply a standard continuum framework to them [I]; the coupling constant Γ_1 and Hypothesis H1 itself are assumptions of this work [H].

2.5 Substitutional energy, E_{sub}

The chemical cost of exchanging a host cation for a substituent, referenced to external reservoirs, is written in the standard point-defect form

$$E_{sub} = \sum_{\alpha} N_{\alpha} \left(\Delta E_{\alpha}^f - \mu_{\alpha} \right) + \sum_{\langle ij \rangle} J_{\sigma_i \sigma_j}, \quad (14)$$

where N_{α} is the number of sites occupied by species α and μ_{α} the corresponding chemical potential imposed by the ore fluid. The formation energies are defined so as to interface directly with supercell DFT (Freysoldt et al., 2014),

$$\Delta E_{\alpha}^f = E_{def} - E_{bulk} + \sum_{\beta} n_{\beta} \mu_{\beta} + q(E_F + E_{VBM}) + E_{corr}, \quad (15)$$

with E_{corr} the finite-size electrostatic correction. In the canonical ensemble used here the reservoir terms are constants and do not influence the configurational statistics; Eq. (14) is retained in full because a grand-canonical treatment, in which they control the equilibrium In content, is the natural extension of this work.

The matrix $J_{\alpha\beta}$ carries all short-range chemical interaction *beyond* the electrostatics already contained in E_c — principally Cu 3d–In 5s–S 3p hybridisation, which has no counterpart in a formal-charge description. In the absence of *ab initio* constraint we adopt $J_{Cu-In} = -0.10$ eV and $J_{Cu-Cu} = J_{In-In} = +0.05$ eV, i.e. a modest covalent stabilisation of the heterovalent pair and a modest repulsion between like substituents. Because these are the only genuinely unconstrained interaction parameters in the model, all principal results are reported both with and without them.

Basis: Eqs. (14)–(15) are standard defect-thermodynamic definitions [F]; the values of $J_{\alpha\beta}$ are hypotheses of this work [H].

2.6 Defect term, E_{def}

Cation vacancies are treated on the same footing as substituents,

$$E_{def} = \sum_i \delta_{\sigma_i, \square} \left(E_V^f + \sum_{j \in NN(i)} w_{\sigma_j} \right), \quad (16)$$

where E_V^f is the isolated vacancy formation energy and w_{σ} a short-range association correction. The dominant vacancy–solute association is already captured by E_c through $\delta q_{\square} = -2$, so w_{σ} carries only the non-electrostatic residue and is set to zero at this stage. Vacancy relaxation is represented through an effective negative misfit $\varepsilon_{\square} = -0.20$ entering Eq. (11).

Basis: the electrostatic part of vacancy–solute association follows from E_c [I]; E_V^f , w_{σ} and $\varepsilon_{\square}$ are hypotheses [H].

2.7 Statistical mechanics and order parameters

The configurational partition function and free energy are

$$Z = \sum_{\{\sigma\}} e^{-\beta H[\{\sigma\}]}, \quad F = -k_B T \ln Z = \langle H \rangle - TS_{\text{conf}}, \quad \beta = (k_B T)^{-1}, \quad (17)$$

with the configurational entropy generated implicitly by the sampling rather than approximated in closed form; no mean-field or point-approximation entropy is invoked. The natural dimensionless control parameter is the reduced temperature

$$\Theta \equiv \frac{k_B T}{\lambda}, \quad (18)$$

so that the Cu–In pair binding energy of Eq. (9) corresponds to $\Theta = 2$.

Four order parameters characterise the resulting configurations.

(i) Warren–Cowley short-range order. For species a, b with bulk atom fractions x_b (Cowley, 1950),

$$\alpha_{ab} = 1 - \frac{P_b(a)}{x_b}, \quad (19)$$

where $P_b(a)$ is the probability of finding b in the first coordination shell of a . $\alpha_{ab} = 0$ denotes a random solid solution, $\alpha_{ab} < 0$ association, $\alpha_{ab} > 0$ avoidance; the parameter is bounded below by $1 - 1/x_b$ ($= -49$ at $x_b = 0.02$).

(ii) Pair enrichment factor. The number of Cu–In first-neighbour bonds normalised to the random-solution expectation,

$$R_{ab} = \frac{N_{ab}}{(NZ_1/2) \cdot 2x_a x_b} \quad (a \neq b). \quad (20)$$

Equations (19) and (20) are not independent: substituting $N_{ab} = P_b(a) x_a N Z_1$ gives the identity

$$R_{ab} = 1 - \alpha_{ba}, \quad (21)$$

used as an internal consistency check on the pair statistics.

(iii) Residual charge imbalance. The mean squared tetrahedral charge deviation $\langle \Delta Q_a^2 \rangle$ measures the degree to which local electroneutrality has been achieved. For a random solid solution with $x_{\text{Cu}} = x_{\text{In}} = x$ its value is $4\langle \delta q^2 \rangle = 8x$ exactly, providing a further analytic benchmark.

(iv) Cluster-size distribution. Solute atoms (Cu, In) connected through first-neighbour bonds are grouped by a union–find algorithm, giving the distribution of contiguous cluster sizes. This is the order parameter that distinguishes dispersed solid solution from incipient nanoscale exsolution.

Basis: all definitions and the identity of Eq. (21) are exact [F].

3 Methods

3.1 Lattice representation

Both sublattices are represented on a single integer grid with spacing $a_0/4$ and periodic boundaries, which removes floating-point neighbour searching entirely. Within a cell of edge 4 (grid

units) the four fcc cation basis vectors are $(0, 0, 0)$, $(0, 2, 2)$, $(2, 0, 2)$, $(2, 2, 0)$; the anion generated by cation j sits at $\mathbf{r}_j + (1, 1, 1)$, reproducing the $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4})$ displacement and yielding a one-to-one cation–anion indexing. The twelve first-neighbour vectors are the permutations of $(\pm 2, \pm 2, 0)$, the six second-neighbour vectors the permutations of $(\pm 4, 0, 0)$, and the four cations coordinating anion j are found at offsets $(0, 0, 0)$, $(2, 2, 0)$, $(2, 0, 2)$, $(0, 2, 2)$ from cation j . All adjacency is tabulated once by integer lookup at initialisation. Simulations reported here use $L = 10$ cells per edge, giving $N = 4000$ cation sites and 4000 anion tetrahedra.

3.2 Monte Carlo algorithm

Configurations are sampled by the Metropolis algorithm (Metropolis et al., 1953) in the canonical ensemble, using Kawasaki exchange dynamics (Kawasaki, 1966) in which the species of two sites are interchanged. Composition is thereby conserved exactly, the appropriate constraint for a closed sphalerite grain equilibrating internally after growth.

Proposals are generated by drawing site i uniformly from the list of solute sites S (those with $\sigma \neq \text{Zn}$) and site j uniformly from all N sites. The proposal is symmetric: $|S|$ is conserved under exchange, and if only one of the two sites is a solute the forward and reverse proposal probabilities are both $(|S|N)^{-1}$, while if both are solutes the unordered pair is proposed with probability $2(|S|N)^{-1}$ in either direction. Detailed balance is therefore satisfied by the standard acceptance criterion

$$P_{\text{acc}}(\{\sigma\} \rightarrow \{\sigma'\}) = \min\left[1, e^{-\beta\Delta H}\right]. \quad (22)$$

Restricting i to solutes eliminates Zn–Zn null moves and accelerates sampling by a factor $N/|S| \approx 25$ at 2 at.% substitution, without altering the stationary distribution. The move set connects all configurations at fixed composition through successive solute–host exchanges, so the chain is ergodic in principle; practical limitations at low temperature are quantified below.

3.3 Incremental energy evaluation

Each trial move is evaluated in $O(1)$ operations. Under exchange the single-site terms cancel identically, so only the tetrahedral and pair contributions need be computed. At most eight anion tetrahedra are affected — four at each site, with cancelling contributions when i and j are neighbours and share one — giving

$$\Delta E_c = \lambda \sum_{a \in A(i,j)} \left[(\Delta Q_a + \delta_a)^2 - \Delta Q_a^2 \right], \quad (23)$$

with δ_a the accumulated charge change at tetrahedron a . The pair contributions are accumulated over the $Z_1 + Z_2$ neighbours of each site, excluding the exchanged partner, whose own bond is unchanged. The array $\{\Delta Q_a\}$ and the solute index list are maintained incrementally, and both are re-derived from scratch and compared against their incrementally maintained values after every production run.

3.4 Annealing protocol and equilibration

Configurations are initialised as random solid solutions at fixed composition and cooled through a ladder of 17 temperatures from 3000 K to 320 K, the configuration being carried over between successive stages. Each stage comprises 4000 equilibration sweeps followed by 3000 production sweeps, where one sweep is N attempted exchanges; the full ladder corresponds to 4.8×10^8 trial moves per model variant.

Because the acceptance ratio falls to $\sim 10^{-5}$ below 1300 K, equilibration cannot be assumed and is tested explicitly. Two independent trajectories are propagated at 573 K, one quenched

directly from a random configuration and one pre-annealed through 2000, 1400, 1000 and 800 K, each followed for 1.2×10^4 further sweeps with the order parameter and energy recorded in blocks. For the λ sensitivity scan, three independent seeds per value are run with the same pre-annealing schedule, and the seed-to-seed dispersion is reported as an error bar; this dispersion measures nucleation stochasticity and is the dominant uncertainty in that experiment.

3.5 Parameterisation

Table 1: Model parameters. Energies in eV, lengths in Å.

Symbol	Value	Basis	Source / remark
a_0	5.4093	[F]	ZnS end-member, room temperature
d_{NN}	3.8250	[F]	$a_0/\sqrt{2}$
$\epsilon_0, \epsilon_\infty$	8.3, 5.13	[F]	static / high-frequency, ZnS
K, μ	77, 30.7 GPa	[F]	bulk and VRH shear moduli, ZnS
$r_{\text{Zn}^{2+}}, r_{\text{Cu}^+}, r_{\text{In}^{3+}}$	0.60, 0.60, 0.62	[F]	Shannon radii, CN = IV
λ	0.30 (range 0.227–0.368)	[I]	Eq. (7); scanned over [0, 0.5]
$\epsilon_{\text{Cu}}, \epsilon_{\text{In}}$	0.000, +0.0333	[I]	from Shannon radii
$E_s^{\text{self}}(\text{In})$	0.033	[I]	Eq. (11)
$\epsilon_\square$	−0.20	[H]	effective vacancy relaxation
Γ_1	1.0	[H]	elastic dipole coupling
$J_{\text{Cu-In}}$	−0.10	[H]	covalent stabilisation; results reported with and without
$J_{\text{Cu-Cu}}, J_{\text{In-In}}$	+0.05	[H]	like-solute repulsion
E_V^f	2.0	[H]	placeholder, requires <i>ab initio</i> determination
$x_{\text{Cu}}, x_{\text{In}}$	0.02, 0.02	—	80 atoms each at $N = 4000$

Of the twelve independent quantities entering the Hamiltonian, seven are measured or exactly derived, two are inferred from measured quantities through Eqs. (7) and (11) without adjustable constants, and only three (J , Γ_1 , $\epsilon_\square$, together with the vacancy formation energy) are hypothetical.

3.6 Verification and validation

Because the principal claims are quantitative, the implementation is subjected to a suite of assertions that run before every production calculation. These test, in order:

1. **Topology.** Reciprocity of the cation–cation and cation–anion adjacency tables; the coordination numbers $Z_1 = 12$, $Z_2 = 6$, $\text{CN}(\text{S}) = 4$; and the Lemma of Section 2.2, checked over all twelve neighbours of a reference site.
2. **Energy bookkeeping.** Agreement between the incrementally accumulated energy and a full recomputation after every run; observed deviation 1.8×10^{-14} eV.
3. **Analytic limits.** The values of Section 2.3.4, reproduced to machine precision: isolated Cu and In at 4λ , isolated V_{Zn} at 16λ , the nearest-neighbour Cu–In pair at 6λ , and the binding energy at -2λ .
4. **Ground-state structures.** $E_c = 0$ for the $2\text{Cu}+2\text{In}$ tetrahedral rule, verified for two distinct orderings, with the tetrahedral composition census confirming that all N tetrahedra satisfy the rule.
5. **Random limit.** Over 24 independent random configurations at $N = 6912$, $R_{\text{Cu-In}} = 1.027 \pm 0.029$ (expectation 1) and $\langle \Delta Q_a^2 \rangle = 0.159$ (analytic value $8x = 0.160$).
6. **Order-parameter identity.** Equation (21) satisfied to 2.2×10^{-16} .

7. **Composition conservation.** Solute counts and the solute index list compared against a direct census after every run, guarding against corruption of the proposal distribution under solute–solute exchange.

3.7 Implementation and performance

The model is implemented in Python 3 with NumPy, with the Metropolis kernel and the pair-counting routines just-in-time compiled through Numba. Single-threaded throughput is approximately 5×10^6 trial moves per second at $N = 4000$; the complete set of experiments requires 381 s on one core.

3.8 Simulation matrix

Table 2: Simulation matrix.

Experiment	Purpose	Configuration
A1	Full model	$\lambda = 0.30$, $J \neq 0$; 17-point cooling ladder
A2	Elastic terms only	$\lambda = 0$, $J = 0$; identical ladder
A3	Electrostatics in isolation	$\lambda = 0.30$, $J = 0$; identical ladder
B	Sensitivity to λ	$\lambda \in [0, 0.5]$ eV, $J = 0$, 573 K, 3 seeds
C	Cluster statistics	union–find analysis at 373 K
D	Equilibration test	quenched vs. pre-annealed trajectories at 573 K

Experiment A2 is the decisive control: with the charge-compensation and chemical terms removed the model contains no interaction capable of correlating Cu and In, and must therefore reproduce the random-solution limit. Experiment A3 isolates the contribution of E_c alone and thus tests Hypothesis H1 directly, since any clustering surviving the removal of J is necessarily electrostatic in origin.

4 Results

All results refer to a $10 \times 10 \times 10$ -cell box ($N = 4000$ cation sites) at $x_{\text{Cu}} = x_{\text{In}} = 2$ at.%, cooled through the ladder of Section 3.4. Temperatures are quoted both in kelvin and as the reduced temperature $\Theta = k_B T / \lambda$ of Eq. (18); with $\lambda = 0.30$ eV the hydrothermal ore-forming window of 100–500 °C corresponds to $\Theta = 0.11$ –0.22, and the Cu–In pair binding energy corresponds to $\Theta = 2$.

4.1 Elastic terms alone produce no detectable ordering

Experiment A2 retains the elastic energy of Eqs. (11) and (12) at full strength but sets $\lambda = 0$ and $J_{\alpha\beta} = 0$, and therefore isolates the strain contribution. Across all seventeen temperatures the Warren–Cowley parameter is $\alpha_{\text{In-Cu}} = -0.03 \pm 0.21$ (range -0.56 to $+0.32$; Fig. 1a, grey curve) and the pair enrichment factor is $R_{\text{Cu-In}} = 1.03 \pm 0.21$ (Fig. 1b), both statistically indistinguishable from the random-solution values of 0 and 1. The mean squared tetrahedral charge deviation remains at $\langle \Delta Q_a^2 \rangle = 0.1593 \pm 0.0035$ throughout (Fig. 1c), agreeing with the analytic random-solution value $8x = 0.160$ to better than 0.5%, and the internal energy is constant at $+0.00066$ eV per site — exactly the sum of the Eshelby self-energies of the 80 In atoms, confirming that the elastic pair term contributes nothing measurable.

The reason is quantitative rather than structural. With $\varepsilon_{\text{Cu}} = 0$ and $\varepsilon_{\text{In}} = 0.033$, the largest elastic pair interaction in the model is $\Gamma_1 \varepsilon_{\text{In}}^2 = 1.1$ meV, some forty times smaller than $k_B T$ at

300 °C. Experiment A2 therefore provides direct internal support for Hypothesis H1: in the Cu–In system, size mismatch is not a competitive ordering mechanism, and any correlation between the two elements must originate elsewhere.

4.2 Charge compensation alone generates strong short-range order

Experiment A3 switches on E_c with $\lambda = 0.30$ eV while keeping $J_{\alpha\beta} = 0$, so the only ordering force is the local electroneutrality penalty. The result is a large and unambiguous Cu–In association: at 573 K (300 °C) the model reaches $\alpha_{\text{In-Cu}} = -14.99$ and $R_{\text{Cu-In}} = 15.99$ (Fig. 1a,b, blue curves), i.e. Cu–In nearest-neighbour bonds are sixteen times more abundant than in a random solid solution of the same composition. Simultaneously the residual charge imbalance falls to $\langle \Delta Q_a^2 \rangle = 0.0065$, a 96% reduction relative to the random reference (Fig. 1c).

Adding the hypothetical covalent term (Experiment A1) raises the enrichment further, to $\alpha_{\text{In-Cu}} = -23.48$ and $R_{\text{Cu-In}} = 24.48$ at 573 K (Fig. 1a,b, red curves). The partition of the total effect is thus approximately two thirds electrostatic and one third chemical, and the electrostatic part alone already exceeds an order of magnitude of enrichment. Because λ is fixed by Eq. (7) rather than fitted, this component of the result contains no adjustable parameter.

The magnitude of the driving force explains the robustness of the effect. The electrostatic energy recovered on ordering, $E_c = \lambda N \langle \Delta Q_a^2 \rangle$, falls from 192 eV in the random reference to 7.8 eV in Experiment A3, a recovery of 2.30 eV per In atom — some 47 times $k_B T$ at 300 °C. Local charge compensation is therefore not a perturbation on the configurational statistics of In in sphalerite; it is the dominant term.

Table 3: Order parameters at 573 K (300 °C), $\Theta = 0.165$.

Experiment	λ (eV)	J	$\alpha_{\text{In-Cu}}$	$R_{\text{Cu-In}}$	$\langle \Delta Q_a^2 \rangle$	imbalance removed	H/N (eV)
A1 full	0.30	on	−23.48	24.48	0.0120	92.4%	−0.00528
A3 electrostatic only	0.30	off	−14.99	15.99	0.0065	95.9%	+0.00262
A2 elastic only	0	off	+0.12	0.89	0.1625	—	+0.00066
Random reference	—	—	0	1	0.160	0	—

4.3 The ordering crossover lies far above the ore-forming window

The temperature dependence in Fig. 1a,b is that of a sharp but continuous crossover rather than a gradual drift. In Experiment A3 the enrichment factor rises from 5.4 at 3000 K ($\Theta = 0.86$) through 7.8 at 2400 K ($\Theta = 0.69$) to 15.3 at 2000 K ($\Theta = 0.57$), and is within 5% of its saturated value by 1600 K ($\Theta = 0.46$). Experiment A1, whose additional chemical term deepens the effective pair binding, crosses over some 400–600 K higher. Locating the crossover at $\Theta^* \approx 0.6$ –0.8, i.e. at $k_B T^* \approx 0.3$ –0.4 times the pair binding energy 2λ , is consistent with the mean-field expectation for a system whose ordering is controlled by a single nearest-neighbour bond energy.

The geological consequence is direct and, we suggest, the single most important result of this paper: the entire hydrothermal ore-forming window ($\Theta = 0.11$ –0.22, shaded band in Fig. 1) lies a factor of three to seven *below* the crossover. Over the whole of that window the order parameters are flat to within the resolution of the calculation ($\alpha_{\text{In-Cu}}$ varies by 0.05 between 773 K and 373 K in Experiment A1). Cu–In short-range order in sphalerite is therefore not a low-temperature refinement that switches on at some threshold within the ore system; it is saturated everywhere that sphalerite forms, and its magnitude carries essentially no temperature information.

Fig. 1 Simulated annealing of Cu-In coupled substitution in sphalerite ($L=10$, $N=4000$, $x_{\text{Cu}} = x_{\text{In}} = 2$ at.%)
yellow band = typical hydrothermal ore-forming window (100–500 °C)

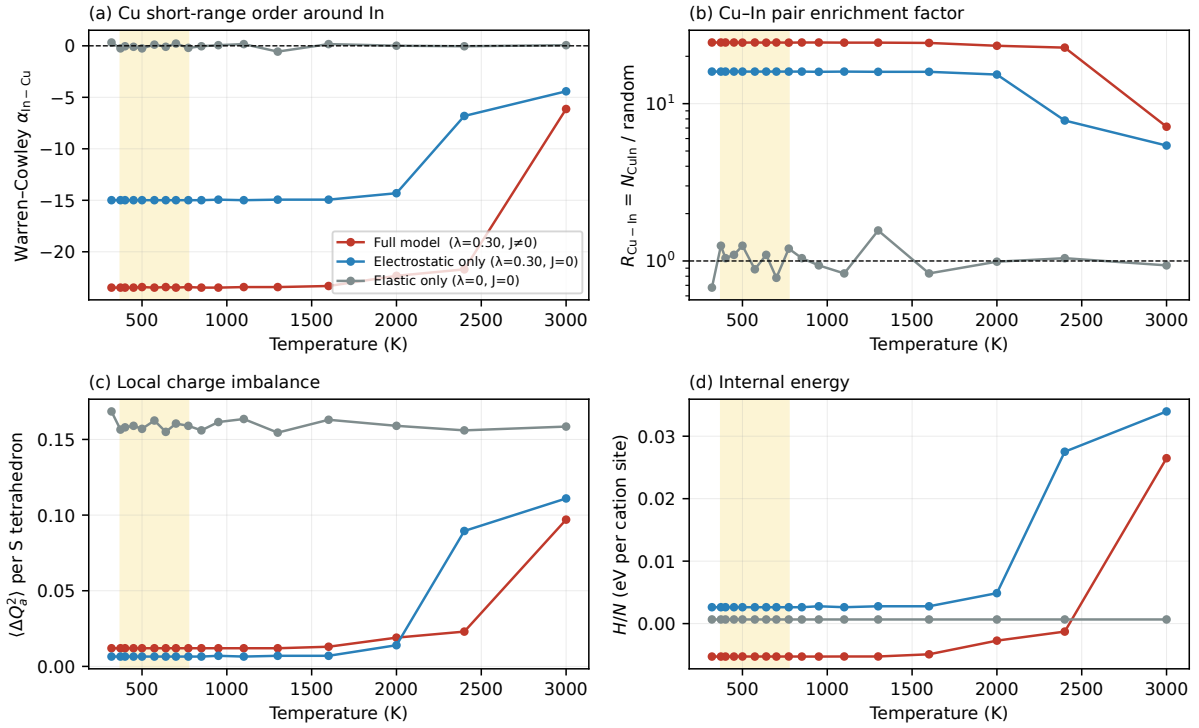


Figure 1: Simulated annealing of Cu-In coupled substitution ($L = 10$, $N = 4000$, $x_{\text{Cu}} = x_{\text{In}} = 2$ at.%). Red, full model ($\lambda = 0.30$ eV, $J \neq 0$); blue, electrostatic only ($\lambda = 0.30$ eV, $J = 0$); grey, elastic only ($\lambda = 0$, $J = 0$). Shaded band, hydrothermal ore-forming window (100–500 °C). (a) Warren–Cowley short-range order parameter $\alpha_{\text{In-Cu}}$; dashed line marks the random-solution value. (b) Cu–In pair enrichment factor $R_{\text{Cu-In}}$, logarithmic scale. (c) Mean squared tetrahedral charge deviation $\langle \Delta Q_a^2 \rangle$; the elastic-only curve coincides with the analytic random value $8x = 0.160$. (d) Internal energy per cation site.

4.4 Robustness to the value of λ

Experiment B scans λ at fixed $T = 573$ K with $J = 0$, three independent seeds per value (Fig. 2). The enrichment factor rises steeply from $R = 1.06 \pm 0.20$ at $\lambda = 0$ through $R = 3.94 \pm 0.20$ at 0.05 eV to $R = 15.68 \pm 0.07$ at 0.10 eV, and is thereafter saturated: all values in $0.10 \leq \lambda \leq 0.50$ eV lie between $R = 15.7$ and $R = 21.1$, with seed-to-seed dispersion (± 3.2 to ± 3.8 for $\lambda \geq 0.15$ eV) exceeding any systematic trend.

Two conclusions follow. First, the dielectric estimate of Eq. (7), $\lambda \in [0.227, 0.368]$ eV (green band in Fig. 2), lies wholly within the saturated regime, so the principal result is insensitive to the choice between the static and high-frequency dielectric constants. It is also insensitive to the neglect of explicit Debye screening: even reducing λ by a factor e^{-1} to 0.083 eV — the most pessimistic correction consistent with $\ell_D \simeq d_{NN}$ — leaves R above 10, and a full order of magnitude of enrichment survives down to $\lambda \approx 0.07$ eV. The qualitative conclusion is therefore stable across roughly a sevenfold range in the only inferred parameter of the model.

Second, and less favourably, the saturation means that the present simulations cannot resolve a threshold in λ , and the large seed-to-seed dispersion above 0.15 eV reflects nucleation stochasticity rather than physics. Both are consequences of the finite box, discussed in Section 5.4.

Fig. 2 Sensitivity to the charge-compensation coefficient λ (electrostatic term only, $J = 0$)

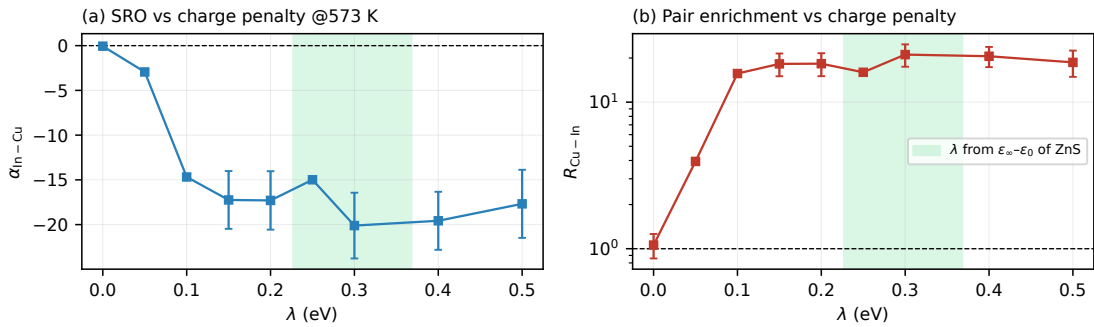


Figure 2: Sensitivity to the charge-compensation coefficient λ at 573 K with $J = 0$; symbols are means of three independent seeds, error bars their standard deviation. Green band, the interval $\lambda \in [0.227, 0.368]$ eV predicted by Eq. (7) from the static and high-frequency dielectric constants of ZnS. (a) $\alpha_{\text{In-Cu}}$. (b) $R_{\text{Cu-In}}$, logarithmic scale.

4.5 Bond-resolved statistics carry a heterovalent signature

The full bond census of the Experiment A1 configuration at 373 K (Table 4) resolves the ordering into its components. Solute-solute bonds of every kind are strongly enriched — the solute sublattice as a whole is aggregated by a factor of 16.8 — but the enrichment is markedly anisotropic with respect to valence. Heterovalent Cu-In bonds are enriched 24.5-fold, whereas homovalent Cu-Cu and In-In bonds are enriched only 9.0- and 9.2-fold. The ratio of heterovalent to homovalent enrichment, 2.70, is the direct fingerprint of E_c : a Cu-In bond carries $\delta q_i \delta q_j = -1$ and is stabilised by 2λ , whereas a Cu-Cu or In-In bond carries $\delta q_i \delta q_j = +1$ and is destabilised by the same amount, so homovalent contacts occur only as a by-product of aggregation rather than as a driving force.

On average each In atom is surrounded by 5.88 Cu atoms out of its twelve cation neighbours, against a random expectation of 0.24.

Table 4: First-neighbour bond census, Experiment A1 at 373 K (24 000 bonds total).

Bond type	Observed	Random expectation	Enrichment
Cu–In	470	19.2	24.5
Cu–Cu	86	9.6	9.0
In–In	88	9.6	9.2
Zn–Cu	318	921.6	0.35
Zn–In	314	921.6	0.34
Zn–Zn	22 724	22 118	1.03
all solute–solute	644	38.4	16.8

4.6 Condensation into a single solute nanodomain

Union–find analysis of the 373 K configurations (Fig. 3a, Table 5) shows that the short-range order is accompanied by complete spatial condensation. In the elastic-only control the 160 solute atoms are dispersed across 120 clusters of mean size 1.33, of which 81% are isolated monomers — the expected percolation statistics of a random 4 at.% solute population on an fcc lattice. In both Experiments A1 and A3 the same 160 atoms form a *single* connected cluster. The corresponding slab projections (Fig. 3b,c) display the contrast directly.

Table 5: Solute cluster statistics at 373 K.

Experiment	clusters	largest	mean size	monomer fraction
A1 full	1	160	160.0	0.00
A3 electrostatic only	1	160	160.0	0.00
A2 elastic only	120	5	1.33	0.81

The internal structure of this domain deserves attention, because it is not the ideal roquesite ordering predicted analytically in Section 2.3.5. Resolving the first coordination shell of In in Experiment A1 gives 0.490 Cu, 0.183 In and 0.327 Zn, against 0.667 Cu, 0.333 In and 0 Zn for bulk chalcopyrite-type ordering. Two distinct discrepancies are present. The Zn fraction of 0.327 reflects the large surface-to-volume ratio of a 160-atom domain and would fall towards zero for larger clusters. The Cu:In ratio within the solute shell, 2.68 against the ideal 2.00, is a genuine departure from the ordered ground state, and is consistent with the residual $\langle \Delta Q_a^2 \rangle = 0.012 > 0$. We therefore describe the simulated object as a *charge-compensated roquesite precursor* rather than as roquesite: the model reproduces the condensation and the heterovalent bonding rule, but at this box size and sampling depth it does not reach the fully ordered $E_c = 0$ state.

4.7 Equilibrium versus kinetic arrest

Because the acceptance ratio falls to $\sim 10^{-5}$ below 1300 K, the plateau of Section 4.3 could in principle be an artefact of configurational freezing rather than a thermodynamic saturation. Experiment D tests this directly (Fig. 4). A trajectory quenched from a random configuration straight to 573 K and a trajectory pre-annealed through 2000, 1400, 1000 and 800 K converge to $\alpha_{\text{In-Cu}} = -23.17$ and -23.48 respectively — a difference of 0.31, or 1.3% of the value — and both order parameters are stationary over the final 1.2×10^4 sweeps. The plateau is therefore thermodynamic in origin.

A residual difference does remain in the energy: the quenched trajectory settles at -0.00211 eV per site against -0.00528 eV for the pre-annealed one (Fig. 4b). The two configurations are thus equivalent at the level of pair statistics but not at the level of total energy, the quenched state lying 3.2 meV per site — 12.7 eV over the box, or 79 meV per solute atom — above the annealed one. This is the model’s internal signature of metastability: the nearest-neighbour

Fig. 3 Emergent Cu-In clustering at 373 K (slab projection along [001])

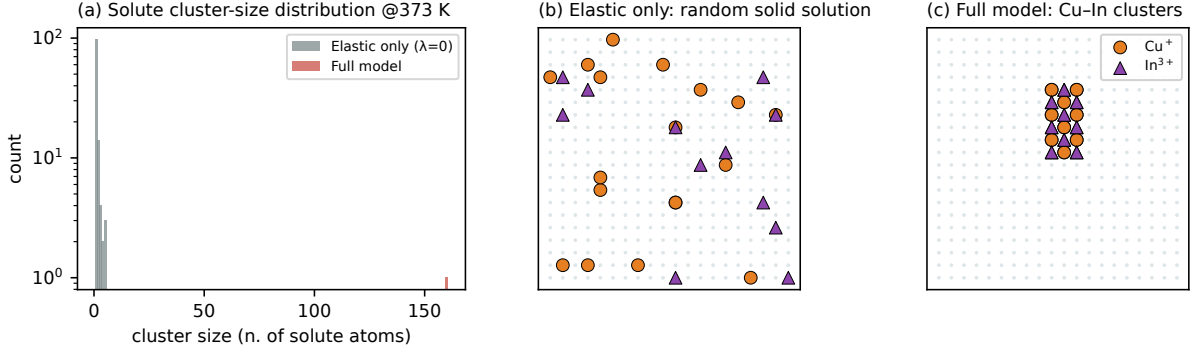


Figure 3: Solute aggregation at 373 K. (a) Cluster-size distribution from union-find analysis of first-neighbour connectivity; grey, elastic-only control (120 clusters, 81% monomers); red, full model (a single 160-atom cluster). (b, c) Projections of a two-cell slab along [001] for the elastic-only control and the full model respectively; small grey symbols, Zn; orange circles, Cu; purple triangles, In.

association equilibrates quickly, while the longer-range optimisation of the domain interior does not.

Fig. 4 Ergodicity test: two independent starting configurations at 573 K

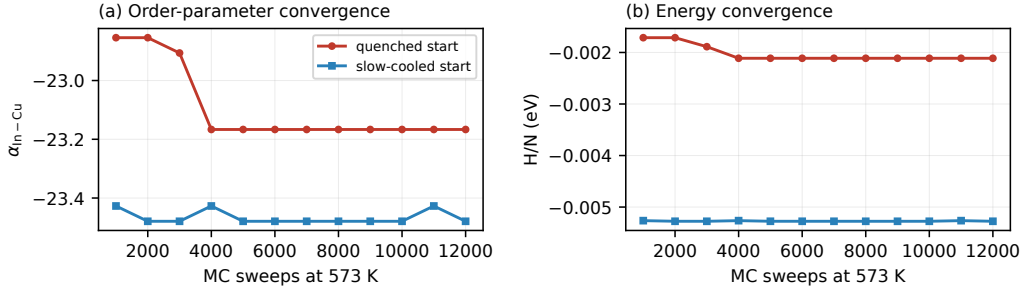


Figure 4: Test of equilibration at 573 K (Experiment D). Red circles, trajectory quenched directly from a random configuration; blue squares, trajectory pre-annealed through 2000, 1400, 1000 and 800 K. (a) Convergence of $\alpha_{\text{In-Cu}}$; the two trajectories agree to 1.3%. (b) Convergence of the internal energy; the quenched state remains 3.2 meV per site above the annealed one, the model's signature of configurational metastability.

5 Discussion

5.1 The Cu-In correlation as a thermodynamic necessity

The near-unity Cu:In molar ratio recorded by LA-ICP-MS in In-bearing sphalerite from a wide range of deposit types is conventionally read as evidence for the coupled substitution of Eq. (1), with the coupling itself treated as an empirical observation to be recorded rather than a quantity to be derived. The results above allow a stronger statement. Starting from the zinc-blende topology, formal ionic charges, and the measured dielectric constant of ZnS — with no fitted parameter and no input describing Cu-In affinity — the model yields a pair binding energy of $-2\lambda \approx -0.45$ to -0.73 eV (Eq. (9)), a sixteenfold enrichment of Cu-In nearest-neighbour bonds at 300°C, and a ground state that is exactly the requisite cation ordering. The 1:1

correlation is not a contingent feature of ore-fluid chemistry; it is what local electroneutrality on this particular lattice topology requires.

Three subsidiary observations sharpen the claim. First, the effect is specifically electrostatic: the elastic terms, retained at full strength in Experiment A2, produce no measurable ordering whatever, confirming Hypothesis H1 within the model and predicting that Ag–In coupling, with its twentyfold larger misfit, should behave qualitatively differently. Second, the effect is valence-resolved: heterovalent bonds are enriched 2.7 times more strongly than homovalent ones, which is the signature of $\delta q_i \delta q_j$ and not of generic solute clustering. Third, the effect is saturated across the entire temperature range in which sphalerite forms, so no ore-forming environment is expected to host In as a genuinely random dilute solid solution.

5.2 A diagnostic for the covalent term

The comparison between Experiments A1 and A3 yields an unanticipated and potentially useful result. Adding $J_{\text{Cu-In}} = -0.10$ eV raises the apparent Cu–In enrichment from 16.0 to 24.5 — but it simultaneously *degrades* local electroneutrality, with $\langle \Delta Q_a^2 \rangle$ rising from 0.0065 to 0.0120 (Table 3). The mechanism is transparent once the ground state of Section 2.3.5 is recalled: perfect charge compensation requires each cation to retain four homovalent neighbours out of twelve, and the assumed $J_{\text{Cu-Cu}} = J_{\text{In-In}} = +0.05$ eV penalises exactly those contacts. The In shell composition of Experiment A1 shows the consequence directly, with Cu:In = 2.68 against the ideal 2.00 (Section 4.6).

This provides a discriminant that does not depend on knowing J in advance. The two quantities $R_{\text{Cu-In}}$ and $\langle \Delta Q_a^2 \rangle$ respond to E_c in the same direction but to J in opposite directions. Measurements sensitive to short-range order and to local charge state — EXAFS coordination numbers, XANES edge positions, or the bond-valence sums extractable from high-resolution structural refinement — could therefore constrain the sign and magnitude of the non-electrostatic term without a full *ab initio* calculation.

5.3 Implications for exploration and resource evaluation

We set out four implications, in decreasing order of confidence. All are predictions of a model calibrated at a single composition on a small box, and require empirical validation.

(i) Cu availability, not In availability, should limit In tenor. If In incorporation must be locally charge-balanced, and if the Cu-compensated route is favoured over the vacancy route by $8\lambda \approx 2.4$ eV (Section 2.3.6), then the achievable In content of sphalerite is capped by the supply of a monovalent compensator. The practical corollary is that In prospectivity should be assessed against the *coupled* availability of Cu and In in the mineralising system rather than against In anomalies alone — favouring, for example, settings in which Cu-rich and Zn-rich assemblages are spatially and temporally linked. This reframes an empirical association already noted in deposit-type meta-analyses as a consequence of lattice thermodynamics.

(ii) Vacancy compensation should be diagnostic of Cu-poor systems. Equation (10) makes the competition quantitative. In-rich but Cu-poor sphalerite should therefore be measurably cation-deficient, and should depart from the 1:1 Cu:In line in a specific direction. This is testable with existing EPMA and LA-ICP-MS data provided that stoichiometry is reported, and constitutes the sharpest near-term falsification test of the model.

(iii) Trace-element short-range order is not a geothermometer. Because the order parameters are flat across the entire ore-forming window (Section 4.3), the degree of Cu–In

association recorded by a natural sphalerite carries almost no information about its formation temperature. More seriously, Experiment D shows that the configuration retains a memory of its *cooling path* — quenched and slowly cooled states differ by 79 meV per solute atom while being indistinguishable in pair statistics (Section 4.7). Any attempt to read formation conditions from trace-element site occupancy must therefore contend with a configurational closure temperature distinct from the growth temperature, and pair statistics alone will not resolve it.

(iv) Spatial heterogeneity should be intrinsic, not sampling noise. The condensation of all solutes into a single domain (Section 4.6) is a nucleation-controlled process, and the large seed-to-seed dispersion of Experiment B is its statistical expression. If this behaviour survives at realistic length scales — which the present box cannot establish — then In should be distributed heterogeneously within individual sphalerite grains, and bulk assay values would average over a distribution rather than measure a representative concentration. The practical implication is that micro-analytical mapping is not merely a refinement of bulk assay for In resources but measures a different quantity. We flag this as the most speculative of the four.

A further consequence concerns the thermodynamic formalism itself. Partition coefficients for In between sphalerite and hydrothermal fluid are ordinarily modelled assuming ideal dilute mixing, i.e. full configurational entropy $-k_B \sum x_\alpha \ln x_\alpha$. The enrichment factors reported here imply that this entropy is substantially overestimated at ore-forming temperatures, and hence that the In solubility inferred from such models is too high. The canonical simulations presented here cannot quantify the correction, because composition is held fixed by construction; doing so requires a grand-canonical treatment.

5.4 Toward the solid-solution versus nanoinclusion boundary

The long-standing question of whether In in sphalerite resides in true solid solution or in nanoscale inclusions is precisely the question that the present calculations *cannot* answer, and it is worth being explicit about why.

At $N = 4000$ and 4 at.% total solute, the 160 solute atoms are sufficient to form exactly one domain of the size that the model wants to build. The system therefore has no way to express an intermediate state: once nucleation occurs, the entire solute budget is consumed, the order parameters saturate, and the distinction between “dispersed pairs”, “several competing nuclei” and “a single inclusion” is lost. This is the origin of the saturation of Fig. 2 above $\lambda = 0.10$ eV and of the seed-to-seed dispersion above 0.15 eV; both are finite-size artefacts and neither should be interpreted physically.

Four requirements follow for any study aiming to settle the question. (a) *Box size.* Resolving a genuine dilute limit alongside competing nuclei requires of order 50^3 cells (5×10^5 sites), which raises the solute population by two orders of magnitude and permits a meaningful cluster-size distribution. (b) *Sampling.* The acceptance collapse to 10^{-5} makes simple Kawasaki dynamics inadequate at ore-forming temperatures; parallel tempering or cluster moves generalising the Wolff construction to the four-state charged model will be required, and 10^8 – 10^{10} trial moves will be needed per state point. (c) *Composition.* A scan across x_{In} from 10^{-3} to 5×10^{-2} is needed to locate the solvus, which is the quantity of economic interest. (d) *Parameterisation.* $J_{\alpha\beta}$, E_V^f and the Fe-dependence of ϵ_r must come from first-principles calculation, since Section 5.2 shows that J controls not only the magnitude of the enrichment but the internal ordering of the resulting domain.

The present paper contributes to that programme in three ways: it supplies the Hamiltonian and its exact analytic limits, it establishes that the electrostatic term alone suffices to drive the transition, and it identifies precisely which observables are and are not resolvable at small box size.

5.5 Limitations

Beyond the finite-size issues above, four limitations should be borne in mind when comparing with natural samples.

First, Fe is absent from the model, whereas natural sphalerite commonly contains up to 10 mol% FeS. Fe^{2+} is isovalent with Zn^{2+} and so does not enter E_c directly, but Fe 3d states modify both the dielectric response that fixes λ through Eq. (7) and the substitution energies of Eq. (15). The sign of the effect is not obvious *a priori*.

Second, the treatment is a rigid-lattice one. Atomic relaxation is represented only through the continuum expressions of Eqs. (11) and (12), which are in any case of marginal validity for large misfits; the Eshelby form should not be applied to Ag^+ ($\varepsilon = 0.67$) without independent verification, despite the comparison drawn above.

Third, Monte Carlo sweeps are not physical time. The freezing documented in Section 4.7 is a property of the sampling algorithm, and although we have argued that it has a natural geological analogue in configurational closure, the model as formulated makes no kinetic predictions and cannot be used to estimate diffusion-limited closure temperatures.

Fourth, all calculations are at a single composition, $x_{\text{Cu}} = x_{\text{In}} = 2$ at.%, which lies at the upper end of natural In contents. Behaviour in the ppm range — where the great majority of economically relevant sphalerite sits — has not been sampled.

6 Conclusions

1. An exact topological property of the zinc-blende structure — that two nearest-neighbour cations share precisely one bridging anion — allows a local electroneutrality functional on the sulphur tetrahedra to be reduced without approximation to a nearest-neighbour pair interaction on the cation sublattice, Eq. (6).
2. The coefficient of that interaction is fixed by the dielectric constant of ZnS rather than fitted, $\lambda \in [0.227, 0.368]$ eV, giving a Cu–In pair binding energy of $-2\lambda \approx -0.45$ to -0.73 eV.
3. The ground state of the charge-compensation term at Cu:In = 1:1 is exactly the roquesite (CuInS_2) cation ordering, recovered without any structural input describing that phase. The ground state is degenerate across orderings satisfying the two-Cu-two-In tetrahedral rule, so polytype selection requires the chemical and elastic terms.
4. Monte Carlo simulation confirms these predictions. Electrostatics alone enriches Cu–In nearest-neighbour bonds sixteenfold at 300 °C and removes 96% of the local charge imbalance; heterovalent bonds are enriched 2.7 times more strongly than homovalent ones. Elastic terms, retained at full strength in a control, produce no detectable ordering.
5. The ordering crossover lies at $\Theta^* \approx 0.6\text{--}0.8$, a factor of three to seven above the entire hydrothermal window. Cu–In short-range order is therefore saturated wherever sphalerite forms and carries no thermometric information.
6. The Cu-compensated route is favoured over vacancy compensation by $8\lambda \approx 2.4$ eV, giving the falsifiable criterion of Eq. (10): vacancy compensation should be confined to Cu-poor systems and detectable as cation deficiency.
7. Finite box size prevents resolution of the solid-solution versus nanoinclusion boundary. Settling that question requires 50^3 -cell boxes, enhanced sampling, a composition scan, and first-principles parameterisation of the chemical term.

Data and code availability

The complete source code, the numerical results underlying all figures and tables, and the scripts that regenerate every figure from scratch are openly available at <https://github.com/Ruqing1963/sphalerite-lattice-mc>. The repository includes the verification and validation suite of Section 3.6, which reproduces every analytic limit quoted in this paper. Code is released under the MIT licence and data under CC BY 4.0.

Author contributions

He Yu: conceptualisation, geological framing, validation, writing — review and editing. **Ruqing Chen:** methodology, formal analysis, software, investigation, visualisation, writing — original draft. **Huanzhang Lu:** critical review of the full manuscript, key revision suggestions, and participation in multiple rounds of revision.

Declaration of competing interests

The authors declare no competing financial or personal interests.

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